VESSEL TRAYS AND ASSOCIATED HARDWARE

FWEL CONTRACT No : **1-14-16601/19**

CONTRACT NO. : **POGC-760-81-156**

PROJECT : **SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)**

COMPANY : **NATIONAL IRANIAN OIL COMPANY**

SITE : **ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN**

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This document is the property of N.I.O.C. Any unauthorised attempt to reproduce it, in any form, is strictly prohibited.

1. SCOPE

This Specification covers the minimum requirements for the design, material selection, fabrication, assembly, inspection, testing and shipment of vessel trays and associated hardware for SOUTH PARS PHASES 9 & 10 Project.

2. REFERENCE DOCUMENTS

This document must be read in conjunction with the following specifications.
Codes and Regulations and Industry Standards which, by being referenced, shall be a part of this Specification. Local codes and regulations shall take precedence when they have more stringent requirements. Latest edition at the time of contract award shall be used. When a conflict exists between the various documents, the more stringent requirement shall govern. Compliance with Contractor / Company specifications and standards will not relieve the manufacturer of his guarantees or other obligations.

2.1 Project Specification

DB 6340 S 999 P 332 204	Basic engineering design data

2.2 Construction Specification

PS TCS 201	Pressure vessel tolerances

3. CODES AND STANDARDS

3.1 Design code and Standards

(ASME) Boiler and Pressure Vessel Code:	
Section II Parts A, B & C	Material Specifications
Section II Part D	Material Properties
Section VIII Division 1	Pressure Vessels
Section VIII Division 2	Pressure Vessels-Alternative Rules
Section IX	Welding and Brazing Qualifications
ASME B1.1	"Unified Screw Threads"
ASME B16.5	"Steel Pipe Flanges and Flanged Fittings"
ASME B46.1	"Surface Texture" (Surface roughness, waviness and lay)
(ASTM) American Society of Testing Materials	"Material Specification"

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4. DOCUMENT PRECEDENCE

It shall be the manufacturer's responsibility to be knowledgeable of the requirements of the referenced Codes and Standards.

The manufacturer shall notify the contractor of any conflict between this Specification, the project tray data sheets, the Codes and Standards and any other specifications noted herein. Resolution and/or interpretation precedence shall be obtained from the contractor in writing before proceeding with the design/manufacture.

In case of conflict, the order of precedence shall be :

- Tray Data Sheet(s)
- Tray Specification
- Project Specifications and Standards
- Codes and Standards

5. SPECIFICATION DEVIATION/CONCESSION CONTROL

Any technical deviations to the Purchase Order and its attachments including, but not limited to, the project tray data sheets and Tray Specification shall be sought by the manufacturer only through Concession Request format. Concession Requests require contractor's and company's review/ approval prior to the proposed technical changes being implemented. Technical changes implemented prior to contractor/company approval are subject to rejection.

6. QUALITY ASSURANCE/QUALITY CONTROL

The manufacturer's proposed quality system shall fully satisfy all the requirements of ISO 9001-2000.
The quality system shall provide for the planned and systematic control of all quality related activities performed during design/development, production, installation or servicing (as appropriate to the given system). Implementation of the system shall be in accordance with the contractor's Quality Manual and Project Specific Quality Plan, which shall both together with all related/reference procedures be submitted to the contractor/company for review and approval as required by the Purchase Order documents.

7. DOCUMENTATION

The manufacturer shall submit the type and quantity of drawings and documentation for contractor's approval or information as listed in the individual Material Requisitions and Purchase Orders.
Mutual agreement on scheduled submittal of drawings and engineering data shall be an integral part of any formal Purchase Order.

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Comments made by the contractor on drawing submittals shall not relieve the manufacturer or sub-manufacturer's of any responsibility in meeting the requirements of the specifications. Such comments shall not be construed as permission to deviate from requirements of the Purchase Order unless specific and mutual agreement is reached and confirmed in writing.

Each drawing shall be provided with a title block in the bottom right-hand corner incorporating the following information:

- Official trade name of the company.
- Manufacturer name and drawing number.
- Drawing title giving the description of contents whereby the drawing can be identified.
- A symbol or letter indicating the latest issue or revision.
- Purchase Order number and item tag numbers.

Revisions to a drawing shall be identified with symbols adjacent to the alterations, a brief description in tabular form of each revision shall be given and, if applicable, the authority and date of the revision shall be listed. The term "Latest Revision" shall not be used.

8. SUB-MANUFACTURERS

The manufacturer shall assume process and mechanical responsibility and overall guarantee for the tray package and associated hardware.

The manufacturer shall transmit all relevant Purchase Order documents including specifications to his sub-manufacturers.

It is the manufacturer responsibility to enforce all Purchase Order and specification requirements on his sub-manufacturers.

The manufacturer shall submit all relevant sub-manufacturers drawings and engineering data to the contractor.

The manufacturer shall obtain and transmit all sub-manufacturers warranties to the contractor/ company, in addition to the system warranty.

9. SHIPPING AND HANDLING

9.1 PACKAGING AND SHIPPING

Preparation for shipment shall be in accordance with the manufacturer's standards and as noted herein. The manufacturer shall be solely responsible for the adequacy of the preparation for shipment with respect to materials and application, and shall provide the tray package and associated hardware at the destination in ex-works condition.
Adequate protection shall be provided to prevent mechanical damage and atmospheric corrosion in transit and at the job site.

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Preparation for shipment and packing will be subject to inspection and rejection by contractor's/ company's inspectors. All costs occasioned by such rejection shall be to the account of the manufacturer.

The trays shall be boxed or crated to prevent damage during transit, lifting or unloading.

All separate and loose items and all spare parts shall be completely boxed or crated. All boxed or crated items shall be identified by item number and service and marked with contractor's Purchase Order number, tag number, and weight, both inside and outside of each individual box or crate. A bill of materials shall be enclosed in each box or crate.

All surfaces, including bolting, shall be given a coating of corrosion inhibiting compound. The corrosion preventative applied shall be indicated on all boxes or crates by manufacture and type.
One complete set of the installation, operation, and maintenance instructions, shall be packed in the boxes or crates. This is in addition to the number called for in the Purchase Order.

9.2 PRESERVATION AND STORAGE

The trays and associated hardware shall be protected to withstand ocean transit and an extended period of storage at the job site. The extended period shall be 18 months minimum. The trays shall be protected against all adverse environments, such as : humidity, moisture, rain, dust, dirt, sand, mud, salt air, salt spray and sea water.

All trays and associated hardware shall be preserved and export packed in accordance with Instructions to be issued by POGC.

10. RESPONSIBILITIES

10.1 MANUFACTURER RESPONSIBILITIES

10.1.1 The tray manufacturer shall guarantee that the trays and associated hardware meet the operating conditions specified. If the trays do not fulfil the performance guarantee, the tray manufacturer shall, without cost to contractor/company, furnish new hardware or redesign and rebuild the furnished hardware to meet the performance guarantee.

The tray supplier is responsible for ensuring that the mechanical design of the trays including the support of the trays is satisfactory for the design conditions indicated on the requisition and/or drawings. Calculations and thickness of material supplied to the tray manufacturer are for information and tendering purposes only. The tray manufacturer shall make his own calculations for which he is fully responsible.

10.1.2 Leakage rates, when specified, shall be certified by the tray manufacturer.

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10.2 SUBMITTALS

10.2.1 General

a. All documents shall be in reproducible form and the quality shall allow for microfilming without loss of clarity. Each document shall indicate that it has been checked before submittal.
Fabrication shall not proceed until approval has been received from the contractor.

b. Review of tray manufacturer documents by the contractor does not relieve the tray manufacturer of responsibility for compliance to the applicable Codes, Standards, project tray data sheets and Specifications.

c. The tray manufacturer drawings, calculations and engineering data, shall be in English language and in metric units.

d Units of measurement shall be compatible with project tray data sheets/drawings and project specifications.

10.2.2 Tray Manufacturer

The tray manufacturer shall submit to the contractor the following engineering data for review :

a. Tray performance simulations.

b. All drawings required for the fabrication and assembly of the trays.

c. Detail drawings of tray support rings, downcomer bars, beam supports, internal distributors and all other tray support parts, including required weld sizes.

d. Complete installation instructions and parts list.

e. List of all components and their ASME or ASTM material specification and grade.

f. For heavy loaded beam support clips the corresponding loads shall be specified by tray manufacturer to enable the vessel manufacturer to check the feasibility to weld directly to the alloy cladding or overlay for carbon steel vessels. If this is impossible, the welding shall be performed on the base material (carbon steel), completed with welding alloy deposit restoration."

g. Copies of material certificates.

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10.2.3 Vessel Manufacturer

The vessel manufacturer shall provide tray support detail drawings or the tray manufacturer drawings identified with the vessel manufacturer's drawing number.

10.3 Tolerances

10.3.1 Tray Supports

Dimensional tolerances shall apply to the completed vessel and shall conform to standard drawing PS TCS 201 and shall be confirmed by the tray manufacturer.

10.3.2 Tray Assembly

Dimensional tolerances shall apply to the assembled trays and shall be given by the tray manufacturer.

10.4 Supplied by tray/vessel manufacturers

10.4.1 Tray manufacturer

a. Tray designs shall be based on the process and mechanical design conditions specified on project tray data sheets/drawings, and this Specification. Tray diameters and number of passes specified are a preliminary selection and alternatives shall be proposed if required to meet the performance guarantee.

b. Tray assemblies shall include all component parts such as, but not limited to, tray decks, weirs, downcomers, insulating baffles, drawoffs, troughs, seal pans, valves, risers, bubble caps, beams, trusses, clamps , bolts, nuts, washers, gaskets, and packing.

c. The following extra hardware shall be supplied with each tray package:
- Bolts and Nuts : 10 percent extra or 10 of each type.
- Washers : 10 percent extra or 10 of each type.
- Tray Clamps : 10 percent extra or 5 pieces.
- Gaskets and Packing Rope : 25 percent extra or 10 feet.

d. The tray manufacturer shall design the support rings, downcomer bars, beam support clips and all other tray support items welded to the vessel.

e. The tray manufacturer shall design all fixed and removable internal pipe distributors and baffles.

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10.4.2 Vessel Supplier

a. The vessel manufacturer shall fabricate and provide tray support bars, downcomer bars, beam support clips, fixed and removable pipe distributors/baffles, in accordance with the tray manufacturer's drawings and this Specification.

b. When specified in the Purchase Order, installation of the tray assemblies and distributors/ baffles, in accordance with the tray manufacturer's drawings and this Specification.

c. If the tray assemblies are specified to be shop installed, the vessel manufacturer shall design the vessel for shipment and erection with the additional tray weight included.

11. DESIGN

11.1 Design requirements

11.1.1 Design criteria

a. General

- The tray manufacturer shall allow for manufacturing tolerances on both the vessel and trays. Specific allowance shall be made for vessel shell out-of-roundness permitted by the ASME code. Allowance shall be made for differential thermal expansion between the vessel shell and the trays during normal startup and operation and normal and emergency shutdown.

- Tray parts (except parts welded to the vessel) affected by tolerance dimensions specified in standard drawing PS-TCS-201, and per tray manufacturer requirements, shall be made adjustable, so that when installed, the tray parts shall be within the specified tolerances.

- Like parts of tray assemblies shall be made interchangeable. Symmetrical tray layouts shall be used wherever possible.

- Tray parts shall be removable through the vessel manholes. Exceptions to this requirement are large major beams and trusses which may be installed before completion of vessel. Contractor approval is required.

- The weight of any single piece of removable tray shall not exceed the following criteria :

  35 kgs when vessel diameter is 1.2 m or less.

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70 kgs when vessel diameter is over 1.2 m.

Removable manhole panels shall not weigh more than 30 Kgs.

- Tray and downcomer sections shall be securely clamped or other quick fastening system to the support rings, bars, and beams/truss members. Bolted joints should be avoided except for the attachment of support members to support clips, attachment of downcomer plates and attachment of auxiliary internals. When the connection of a support member to a clip is essential to the integrity of the tray construction, the bolted connection shall be secured by means of tack welding the nut to the bolt, or by employing a second nut. Where required, joints shall be sealed with tray manufacturer supplied gaskets. Where tray vibration is expected bolted-through tray connections should be used along with lock washers or self locking nuts rather than frictioned fastening.

- All trays shall be provided with at least one removable manhole. Single pass trays shall be provided with one manhole, several pass trays shall be provided with the same number of manholes as passes located on each side of the central down-comers. The tray manholes shall be vertically aligned and located in the valve seating area. Small tray sections may be used for a manhole. Tray manholes shall have a minimum opening of 320 x 500 mm. Where this is impossible for construction reasons, the use of smaller manholes shall be subject to the approval of the contractor/company. The manhole shall never be smaller than270 x 450 mm. If the clearance between the tray deck and the bottom of major supporting beams above is less than 400 mm, a tray manhole shall be provided on each side of the beams. The tray manufacturer shall ensure that passage of a man is possible through any tray manhole in the installed position.

- Tray spacing shall be as specified on Project tray data sheets/drawings.

- Major support beams shall be parallel to the flow of liquid. The total height of a support beam shall not exceed one half of the tray spacing. On conventional downcomer trays where beams are perpendicular to the direction of liquid flow, the beam shall occupy not more than one third of the space between the trays, as defined by:

  A_{beam} /(D x TS) < 1/3
  where A_{beam} = Area of beam, horizontally projected, m2
  D = Internal diameter of column, m
  TS = Tray spacing, m

- Unless otherwise noted on the project tray data sheets/drawings, small projections for valve trays shall be provided on the valve seating surfaces to prevent sticking of the valves to the valve seats. The guide legs and vertical

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travel stops shall be arranged so the valves cannot work through the holes during normal operation. An antirotation device shall be provided to prevent spinning of the valves.

- The net free area of a tray is defined as the ratio of the total hole area to the column cross-sectional area expressed as a percentage. Maximum allowable deviation of the net free area shall be plus or minus 0.5% of the column cross sectional area, or plus or minus 5.0% of the stated net free area, whichever is the smaller. The hole area specified does not include an allowance for any holes which may be blocked by support members. The manufacturer shall ensure that the unobstructed hole area is as specified.

- The perforating pattern may be rectangular or triangular. The pitch of perforations shall not be less than 1.5 times the diameter of the perforation.

- Stacked cartridge trays shall be installed in vessels when the nominal diameter is smaller than 800 mm. Stacked trays shall be designed to permit installation and removal through the vessel top. Vessels with diameters 800 mm or larger shall have tray support rings welded to the vessel shell.

- Cartridge and pedestal type trays shall be designed to minimize edge leakage.

- Unless otherwise specified, all trays shall be designed to permit installation and removal from the top side of the tray.

- Except for total draw-off trays, at least one 13 mm diameter drain hole shall be provided on each tray or seal pan to permit draining of the liquid. Drain holes in recessed seal pans shall be located downstream or on the tray side of the downcomer so any vapor passing through the holes will not by-pass the tray above. Inlet weirs shall have a 13 mm hole at the base to permit drainage into the tray. Drain holes shall not be provided in total draw-off trays.

b. Minimum Thickness and Corrosion Allowance

- Thickness of the tray parts shall be determined by calculation based upon the design loads. A corrosion allowance if any, shall be added to this thickness. However, total thickness including corrosion allowance shall not be less than as listed in Attachment 1, Table1.

- Thickness and width of tray support rings, down-comer bolting bars and supporting beam clips shall be determined by tray manufacturer and shall include twice the value of the indicated vessel corrosion allowance unless otherwise stated. Corrosion allowance shall be added to throat dimension of all welds except those joining corrosion resistant materials

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- When independent carbon steel beams are designed by tray manufacturer, calculations shall assume that beam thickness has been reduced by one value of the indicated vessel corrosion allowance.

- The minimum corrosion allowance for removable tray parts shall be as follows :

  -Alloy material None
  -Carbon steel 0.76 mm total

c. Loads and Deflections

- Design Loads

  – All trays, pans, and supports shall be designed for the corroded thickness at design temperature.

  – Tray decks and seal pans shall be designed for the greater of a uniform live load of 200 kg/m2 or dead load plus liquid height of 50 mm above the weir setting.

  – Areas directly below downcomers shall be designed for the greater of 300 kg/m 2 uniform live load or dead load plus liquid height of 450 mm, whichever is greater.

  – Shed rows, disc-and-donut types of trays with no normal operating liquid level shall be de-signed for a uniform live load of 120 kg/m2 on the projected horizontal area plus dead load.

  – Frequent pressure surges, such as those occurring when slugs of water are introduced into a hot column or into circulating reflux streams, may cause tray damage unless“explosion proof” trays are used. Such trays are designed to withstand a minimum load of 3000 kg/m2 from either top or bottom without yielding.

  – Support beams and beam clips shall be designed for the total dead and live loads of the supported parts.

  – For maintenance purposes, tray decks and seal pans, together with their support mem-bers, shall be designed for a concentrated live load of 110 kg at any location, independent of other design live loads.

  – The allowable stresses for structural members (support rings, bolting bars, beams, beam clips, etc.) shall be in accordance with Section II, Part D of the ASME CODE.

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- Deflections
  - The calculated combined deflection of a tray deck and support beam/truss due to operating loads shall not exceed 1/900 times vessel ID or 5 mm whichever is less.
  - Support beams or trusses may be cambered subject to contractor's approval.
- Levelness
  - The maximum tilt of a tray deck shall be same as followings indicated on TOTAL standard number PS. TCS. 201.

Tower diameter (mm)	Tray levelness* (mm) (high to low)
1200 and under	3.0
1201 to 2100	5.0
2101 to 3900	6.5
3901 to 4800	8.0
Over 4800 mm	9.5

  - Local depressions and bulges in a tray panel shall not exceed 6 mm from nominal level.

d. To facilitate access to the trays and other internals, the following location of column manholes is recommended :

- Every 15 trays unless otherwise stated on the project tray data sheets/drawings.
- Next to gas or liquid inlet devices.
- Below total draw-off trays.
- In the bottom compartment.
- Top of the column.
- Directly above draw-off trays

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11.1.2 Basic Materials and Methods

a. Tray Supports

- All materials for tray support rings and other parts welded to the vessel shall be as specified on the project tray data sheets/drawings.

- Slotted holes shall be provided for all support clips welded to the vessel. Size and quantity of slots shall be determined by the tray manufacturer.

b. Trays, Valves, and Hardware

All materials shall be as specified on the project tray data sheets/drawings, and per Attachment1, Table 2 of this Specification.

c. Internal Distributors

Internal distribution pipes shall have flanged connections. Pipes shall be removable through vessel manholes.

d. Gaskets

- When required, gaskets and sealing washers shall be 3 mm minimum thickness.
- Flat gaskets shall be woven tape and packing rope shall be braided, fabricated from asbestos free materials and suitable for the process fluid and temperature. Proposed materials are subject to contractor's approval.

e. Bolting

Alloy bolting shall be 10 mm minimum diameter and carbon bolting shall be 12 mm minimum diameter. Bolt length shall allow for 2-3 threads exposed after installation of nut(s).

f. Substitution

Substitution of materials must have contractor's written approval before fabrication.

11.1.3 Fabrication and Manufacture

a. Welding

- Bottom. The top weld shall be designed as a strength weld, with the bottom a seal weld.

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- Downcomer support bars shall be continuously welded to the shell from each side. Downcomer bars may be welded from one side, only if the other side is not accessible.
- For carbon steel vessels with alloy cladding or overlay, the tray support rings and beam support clips may be welded directly to the carbon steel base metal. Downcomer bars, tray support rings, shall be welded to the vessel with continuous welds both top and bottom and may be welded to the cladding. The area where the down-comer bar will be welded shall be ultra-sonically tested to ensure the integrity of cladding bond before welding.
- Welder's, welding operators, welding procedures and NDE testing procedures shall be subject to approval by the contractor.
- Welding procedures for welding all tray parts shall be submitted to the contractor for review.

b. Workmanship

- Surfaces of fabricated parts shall be smooth and free from scale, dents, and other defects.
- Perforations for sieve trays shall be punched downwards from the top sides of trays.
- Punched valve tray perforations shall be burr side up. Punched holes for snap-in valves shall be burr side down, to facilitate mounting of the valves.
- Exposed edges of trays at locations where personnel may pass shall be beveled or rounded off.
- Burrs shall be removed from perforated areas of trays.
- If nuts are welded to tray parts, the threads shall be chased after welding.
- When flame or plasma arc cutting of tray panels or other tray components is done on stainless steel or other alloys whose corrosion resistance may be affected by this operation, the tray manufacturer shall make provisions to remove the affected area by grinding.
- The tray manufacturer shall perform cleaning or pickling and passivating operations on tray parts as necessary to assure that surface accumulation of contaminants, due to forming dies will not result in preferential corrosion or hinder the free operations of the units.

11.2 Inspection and tests

11.2.1 General

Tray decks and seal pans shall be subject to inspection by the contractor/company in the shop where they are fabricated. Such inspection shall not relieve the tray manufacturer of any responsibility for the performance guarantee.

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11.2.2 Trial Assembly

One tray assembly from each group of similar trays shall be trial assembled by the tray manufacturer Parts shall be selected at random to ensure fitup. The trial assembly shall be supported in a manner comparable to final installation. The tray manufacturer shall notify the contractor (10) ten working days prior to the time the assembled tray(s) will be ready for inspection.

11.2.3 Leakage Test

a. The following trays or tray parts shall be designed and installed to meet the leakage rates described below :
   - Total draw-off trays over full height.
   - Partial draw-off trays over full height.
   - Bubble cap trays up to weir or riser height.

b. With all drain holes plugged and the water level at the height indicated in paragraph 11.2.3.a, the drop in level shall not exceed 5 cm in 30 min. Leakage through joints of bubble cap trays shall be uniformly distributed across the tray.

c. After testing, all plugs shall be removed.

d. The tray manufacturer shall certify the trays will not exceed the stated leakage rate based upon an actual leakage test stated above. This certification may be based on tests of trays of identical design but not necessarily of the same size.

11.3 Identification and tagging

11.3.1 Tray Parts

Each tray part shall be piece marked for identification as shown on the tray manufacturer's assembly drawings. Only paint or ink which contains non-metallic pigment shall be used for marking. Paint or ink shall not contain substances (e.g. sulphur, chloride, etc) which would be harmful to these materials at ambient or elevated temperatures.

11.3.2 Hardware

Small parts such as bolting, washers, and clips shall be packaged and marked with proper identification.

11.4 Preparation for shipment

11.4.1 All items shall be boxed or crated to suit the particular mode of transportation in order to prevent damage during shipment. All boxes or crates shall be marked with contractor's Purchase Order number, tag number, and tray numbers.

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11.4.2 Each tray assembly shall be packaged as a complete unit.

11.4.3 If required, the tray components shall be packaged for overseas shipment in accordance with special instructions given in the Purchase Order.

11.4.4 Tray parts for different vessels shall not be intermixed.

11.4.5 In addition to the requirements of Section 10.2.2d., of this Specification, a copy of the parts list and installation instructions shall be included in the shipment of trays for each vessel.

12. INSTALLATION

12.1 Tray assemblies shall be installed in accordance with the tray manufacturer's drawings and instructions.

12.2 Seal welding of tray parts, if required, shall be welded from the top side whenever possible. The tray manufacturer shall specify the total weld length size and type.

12.3 In general, side sumps and troughs shall be 100% welded. Field welding draw-off tray parts to each other during site installation is not allowed unless required due to manhole constraints.

12.4 Gasketed joints below normal liquid level shall be avoided wherever possible. Long straight joints below normal liquid level, when they cannot be avoided, shall be seal or strength welded. Facing or circular joints to pipe connections shall be machined (stock-finished).

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ATTACHMENT 1

TRAY PARTS TABLES

Table No. 1
Tray Parts Minimum Thickness
(including corrosion allowance)

Tray Parts	Carbon Steel	13 Cr, 18 Cr - 8 Ni, 16 Cr - 12 Ni - 2 Mo, Alloy 625, Alloy 825
Valves	-	1.50 mm
Bubble, Caps Risers, Hats	3.40 mm	1.50 mm
Decks, Downcomers, Weirs	3.40 mm	2.00 mm
Trusses	8.00 mm	6.00 mm
Beams	8.00 mm	6.00 mm
Support Rings	8.00 mm	6.00 mm

Table No. 2
Tray Parts Specified Materials

Material	Decks, weirs,downcomers trusses,beam,caps, risers hold downs, hats	Valves	Bolts, clamps, fasteners (Note 1)
Carbon Steel	No. 1 Finish Carbon Steel	ASTM 410	Commercial Quality Carbon Steel (Note 2)
13 Cr. Steel	ASTM 410S	ASTM 410	ASTM 410
18 Cr - 8Ni Steel	ASTM 304L	ASTM 304	ASTM 304
16 Cr - 12Ni - 2Mo	ASTM 316L	ASTM 316	ASTM 316
Alloy 625	ASTM B443	ASTM B443	ASTM B443
Alloy 825	ASTM B424	ASTM B424	ASTM B424

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NOTE :

1.Nuts shall be of similar material as the bolts. Proper selection shall be made to minimise galling.

2. Carbon Steel trays shall have ASTM 410 stainless steel bolting for tray manways.

3. All welded parts fabricated from 13 Cr. steel shall be Type 410S or Type 405. Non-welded parts shall be Type 410.

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